

AI-Powered Credit Risk Assessment Model

Shrey Shah | XGBoost trained on 1.3M loans | Basel III & IFRS 9 | Gini: 0.44 | KS: 32

An AI-powered credit risk assessment framework trained on 1.3 million real Lending Club loans. The model evaluates loan applications across three analytical layers — quantitative financials, behavioral patterns, and live macroeconomic conditions — before Claude Sonnet 4 synthesises all inputs into a scored decision with full written rationale, stress test results, and IFRS 9 classification.

MODEL VALIDATION METRICS — Basel II IRB Compliant

0.7205 AUC Score	0.4410 Gini Coefficient	32.0 KS Statistic	1.3M Training Loans	19 Features Used	7-Band PD Calibration
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Basel II IRB minimum: Gini > 0.30 | Good model: Gini 0.40-0.55 | KS > 20 required | This model: Gini 0.44 ✓ KS 32 ✓ — meets institutional lending standards

HOW IT WORKS — 3-LAYER ENSEMBLE + AI

Layer	Name	Wt	What It Does
Layer 1	Quantitative Engine	40%	Calculates PD, LGD, EAD, EL using Basel III ratios. XGBoost-driven PD trained on 1.3M loans. Key ratios: DSCR, Leverage, ICR, Current Ratio, DTI, LTV.
Layer 2	Behavioral Engine	35%	Detects anomalies: utilisation spikes, payment velocity, inquiry clustering, income instability, employment risk. Severity-weighted flags (High/Medium/Low).
Layer 3	Macro Overlay	25%	Live data: Fed rate, yield curve (10Y-2Y), CPI, VIX, unemployment, sector default rates. Adjusts PD for economic cycle (Expansion/Peak/Contraction/Trough).

AI Meta-Aggregator — Claude Sonnet 4: Synthesises all 3 layers, runs stress tests (mild/moderate/severe recession), classifies under IFRS 9 (Stage 1/2/3), and produces a fully explainable credit decision with SHAP-driven audit trail traceable to every input.

KEY RATIOS, EDA FINDINGS & DECISION RULES

Ratio	Formula	Threshold	EDA Finding from 1.3M Loans
DSCR	EBITDA / Annual Debt	> 1.25	Below 1.0 = near-certain default. Every 0.25x drop adds ~4% default risk
DTI	Monthly Debt / Monthly Income	< 40%	Linear relationship — every 5% increase adds 2-3% default rate
LTV	Loan / Collateral Value	< 80%	Above 90% LTV = 2x default rate vs below 60%
Grade	Lender credit assessment	A-G	Master variable: Grade A=6.1% vs Grade G=50.1% default rate
PD	XGBoost on 19 features	< 5%=AAA	Trained on 1.3M loans. Gini 0.44, KS 32, AUC 0.72
LGD	1 - Recovery Rate	—	Unsecured: 92.5% loss. Saudi Nafith salary assignment: ~50% recovery
EL	PD x LGD x EAD	—	Drives loan pricing. Grade G debt consolidation: \$3,500+ avg EL
Inquiries	Hard pulls in 6 months	< 3	0 inquiries=18% default vs 5 inquiries=28.2% — strong behavioral signal

OUTPUT PER ASSESSMENT	FRAMEWORKS & TECH STACK
• Composite score 0-1000 with grade (AAA to D) • Decision: Auto Approve / Approve / Conditional / Manual Review / Decline • XGBoost PD (calibrated, trained on 1.3M real loans) • Suggested rate = Risk-free rate + Credit spread (bps) • IFRS 9 Stage 1/2/3 with ECL provisioning basis • Stress test: Mild / Moderate / Severe / GFC scenarios • SHAP-explainable audit trail per decision	• XGBoost — champion model (Gini 0.44, KS 32, AUC 0.72) • Logistic Regression — baseline interpretable model • SHAP — individual loan explainability • Basel III Pillar 1/2, IFRS 9 ECL staging • CFO Dashboard: vintage, recovery, stress test • Python / Flask / Claude Sonnet 4 / Render • 1.3M Lending Club loans (2007-2018)
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